

Backward Pass — Complete Differentiation

Code → Mathematical Differentiation (Chain Rule) | XOR Neural Network (2 → 5 → 3 → 2)

Code	Math (Differentiation)	Explanation
<code>m = X.shape[0]</code>	<code>m = number of samples</code>	Total training examples (4 for XOR)
Output Layer (Layer 3) — sigmoid activation		
<code>dA3 = self.A3 - Y.T</code>	<code>dL/dA3 = A3 - Y</code>	BCE Loss derivative w.r.t output
<code>dZ3 = dA3 * sigmoid_derivative(self.A3)</code>	<code>dL/dZ3 = dA3 x A3(1-A3)</code>	sigmoid'(a) = a(1-a), chain rule
<code>dW3 = (1/m) * dZ3 @ self.A2.T</code>	<code>dL/dW3 = (1/m) x dZ3 . A2^T</code>	Weight gradient, averaged over m
<code>db3 = (1/m) * np.sum(dZ3, axis=1, keepdims=True)</code>	<code>dL/db3 = (1/m) x sum(dZ3)</code>	Bias gradient, sum over samples
Hidden Layer 2 — tanh activation		
<code>dA2 = self.W3.T @ dZ3</code>	<code>dL/dA2 = W3^T . dZ3</code>	Propagate gradient backward through W3
<code>dZ2 = dA2 * tanh_derivative(self.A2)</code>	<code>dL/dZ2 = dA2 x (1 - A2^2)</code>	tanh'(a) = 1-a^2, chain rule
<code>dW2 = (1/m) * dZ2 @ self.A1.T</code>	<code>dL/dW2 = (1/m) x dZ2 . A1^T</code>	Weight gradient for layer 2
<code>db2 = (1/m) * np.sum(dZ2, axis=1, keepdims=True)</code>	<code>dL/db2 = (1/m) x sum(dZ2)</code>	Bias gradient for layer 2
Hidden Layer 1 — tanh activation		
<code>dA1 = self.W2.T @ dZ2</code>	<code>dL/dA1 = W2^T . dZ2</code>	Propagate gradient backward through W2
<code>dZ1 = dA1 * tanh_derivative(self.A1)</code>	<code>dL/dZ1 = dA1 x (1 - A1^2)</code>	tanh'(a) = 1-a^2, chain rule
<code>dW1 = (1/m) * dZ1 @ X</code>	<code>dL/dW1 = (1/m) x dZ1 . X</code>	Weight gradient for layer 1
<code>db1 = (1/m) * np.sum(dZ1, axis=1, keepdims=True)</code>	<code>dL/db1 = (1/m) x sum(dZ1)</code>	Bias gradient for layer 1
Weight Update — Gradient Descent (Not Differentiation)		
<code>self.W3 -= self.lr * dW3</code>	<code>W3 = W3 - lr x dL/dW3</code>	Update weights using gradient

Chain Rule: $dL/dW = dL/dA \times dA/dZ \times dZ/dW$ | $\text{sigmoid}'(a) = a(1-a)$ | $\text{tanh}'(a) = 1-a^2$ | Krishna Sharma | IIT Delhi Internship